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Lecturer

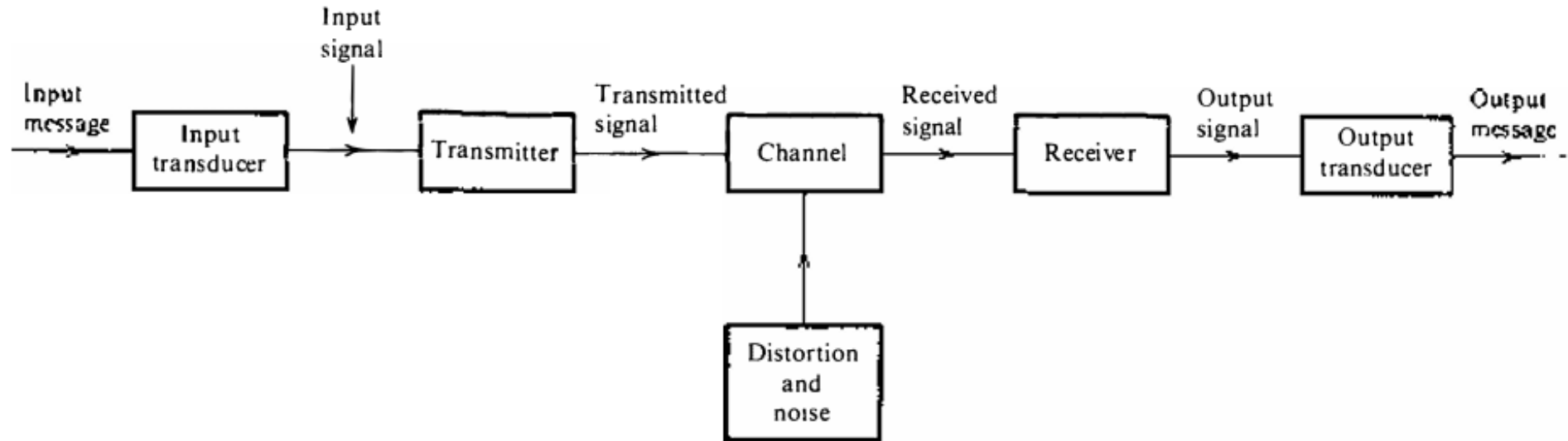
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DIGITAL COMMUNICATION SYSTEM



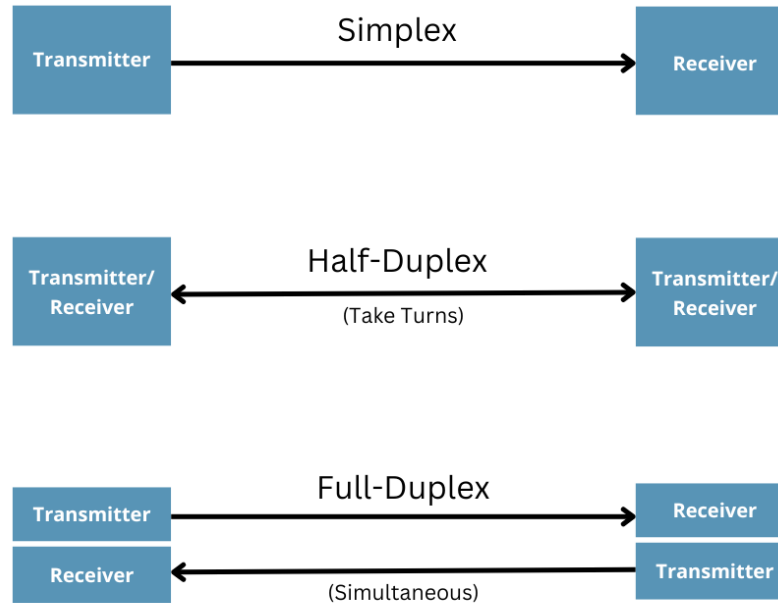
Basic elements of digital communication system:

- **Source**- voice message
- **Input transducer**- physical to electrical
- **Encoder**- error detection, analog to digital
- **Modulator**- make suitable for transmission
- **Medium**- channel like cable, optical fiber
- **Demodulator**- retrieve data from signal
- **Decoder**- remove noise and reconstruct message
- **Output transducer**- electrical signal to sound
- **Destination**- end user

MODES OF COMMUNICATION

Based on data flow:

- **Simplex**- the communication is unidirectional. Example: keyboard, TV, radio
- **Half duplex**- two way but not simultaneous. Example: walkie-talkie
- **Full duplex**- both stations can transmit and receive simultaneously. Example: phone call, video conference



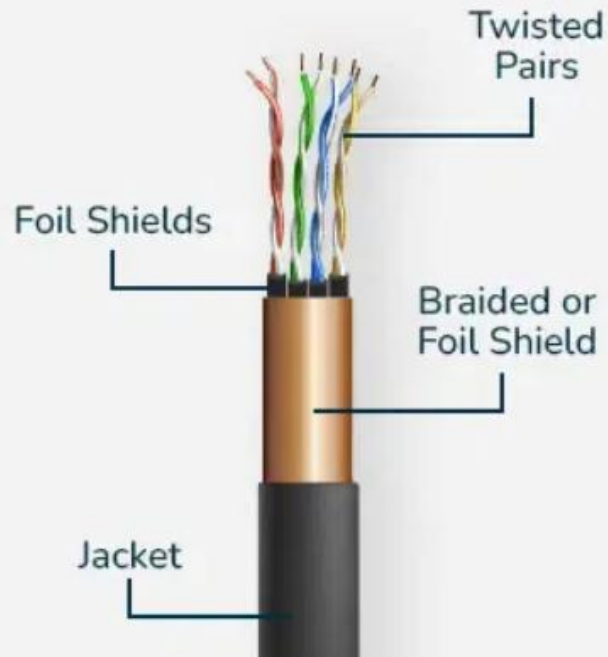
COMMUNICATION CHANNELS

❑ **Guided Communication Channel:** The signal travels through a physical path.

1. **Twisted pair cable:** It consists of two individually insulated copper conductors twisted together in a helical pattern. Popular for short distance and low-cost usage. There are two types: (a) Unshielded & (b) Shielded. Used in LAN, telephone, internet line.
2. **Co-axial cable:** A central copper conductor surrounded by a dielectric insulating layer, a metallic shielding layer, and an outer protective jacket for better protection against noise and interference. Used in cable TV, CCTV,
3. **Optical fiber:** This medium transmits data in the form of light signals through a glass or plastic core using the principle of total internal reflection. It has high band width, low loss and can transmit long distance. Used in medical imaging, internet, submarine communication.

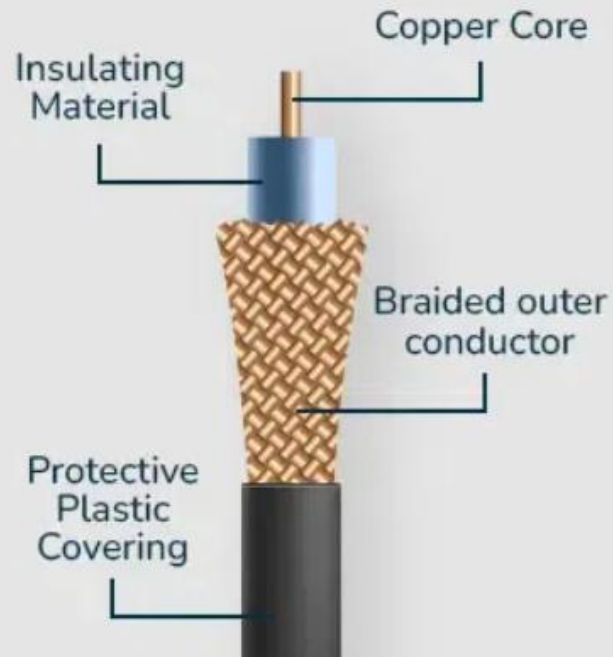
COMMUNICATION CHANNELS

TWISTED PAIR CABLE



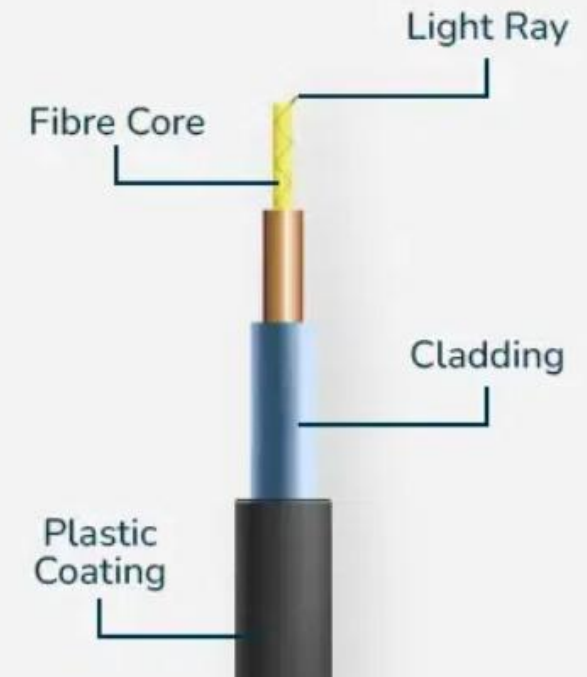
VS

CO-AXIAL CABLE



VS

OPTICAL FIBER

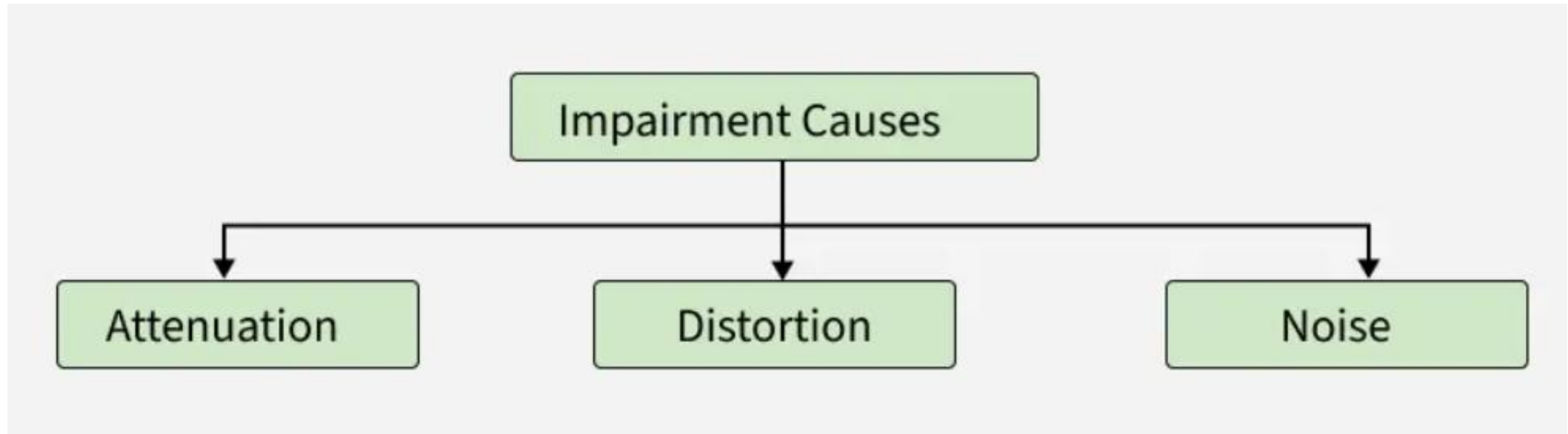


COMMUNICATION CHANNELS

❑ **Unguided Communication Channel:** It uses electromagnetic waves to transmit data without any physical medium.

1. **Radio wave:** It is suitable for broadcasting and wireless communication. Frequency range between 3kHz – 300 GHz. They don't require line of sight between transmitter and receiver. Popular for radio-TV broadcasting, navigation
2. **Microwave:** This media requires line of sight communication hence transmitting and receiving antennas must be properly aligned. Frequency range of 1 GHz to 300 GHz. Applications in cellular communication, satellite.
3. **Infrared:** Infrared mostly used for short-range wireless communication and operate in the frequency range of 300 GHz to 400 THz. They cannot penetrate solid obstacles. Used in sensor automation, TV remote.

TRANSMISSION IMPAIRMENT



NOISY CHANNEL

Noisy channel capacity, also known as Shannon capacity, maximum rate at which information can be transmitted over a noisy communication channel with a small probability of error. The formula states as below:

$$C = B \log_2(1 + SNR)$$

Here,

C = Channel capacity (bits per second, bps)

B = Channel bandwidth (Hz)

SNR = Signal-to-Noise Ratio

If SNR is given in decibel, then

$$SNR = 10^{\frac{SNR_{dB}}{10}}$$

NOISY CHANNEL

Problem 1: Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. For this channel calculate the capacity C .

Problem 2: Signal to noise ratio given in decibel 15 dB and the channel bandwidth is 2 MHz. Find Shannon capacity.

NOISELESS CHANNEL

Nyquist bit rate formula defines the theoretical maximum bit rate for a noiseless channel,

$$\textit{BitRate} = 2B \log_2 L$$

Here,

B = Channel bandwidth (Hz)

L = No. of signal level

BitRate is in bps

Problem 3: Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. What is the maximum bit rate?

Problem 4: We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

REFERENCE

Data Communications and Networking by Behrouz A. Forouzan
Example: 3.34, 3.35, 3.36, 3.37, 3.38, 3.39, 3.40, 3.41



THANK YOU

